



Cairo University, Faculty of Computers
and Artificial Intelligence

FACULTY OF COMPUTERS AND AI, CAIRO UNIVERSITY

First Semester CS213: OOP

Assignment 2 First Phase

Course Instructors:
Dr. Mohamed El-Ramly

Prepared By:

Belal Mohamed Youness	20220087
Mohamed Islam Khaled	20220282
Mohamed Gamal Abd El-Nasser	20221119

Git Hub Screen Shots

problem 10 implemented after several times

Boleka3

committed 7 minutes ago

Verified8c09c1f

<>

Tip

Update A2_S24_20220087_7.cpp

ISAAA7

committed 45 minutes ago

Verifiedc96264d

<>

Tip

Update A2_S24_20220087_4.cpp

ISAAA7

committed 46 minutes ago

Verified9a5097e

<>

Tip

Update A2_S24_20220087_1.cpp

ISAAA7

committed 46 minutes ago

Verifiedfa918cb

<>

Tip

Commits on Oct 30, 2023

problem 7 implemented

Boleka3

committed last week

Verified9ac56d4

<>

Tip

problem 4 implemented

Boleka3

committed last week

Verified7b1b0cd

<>

Tip

Commits on Oct 28, 2023

problem 1 implemented

Boleka3

committed last week

Verifiedf90e853

<>

Tip

Problems added for the least ID

Boleka3

committed last week

Verified0f366ac

<>

Tip

Commits on Oct 27, 2023

Delete A2_S24_20220087

Boleka3

committed last week

Verifiedcaf0e18

<>

Tip

Create A2_S24_20220087

Boleka3

committed last week

Verified6b189a1

<>

Tip

Update A2_S24_20220282_11.cpp

ISAAA7

committed 2 days ago

Verifiede832866

<>

Tip

Commits on Nov 3, 2023

Update A2_S24_20220282_8.cpp

ISAAA7

committed 2 days ago

Verified16797e8

<>

Tip

Update A2_S24_20220282_5.cpp

ISAAA7

committed 2 days ago

Verified6d88b47

<>

Tip

Update A2_S24_20220282_2.cpp

ISAAA7

committed 2 days ago

Verifiedee5d756

<>

Tip

Update A2_S24_20220282_5.cpp

ISAAA7

committed 2 days ago

Verifiedd99ddd1

<>

Tip

Update A2_S24_20220282_8.cpp

ISAAA7

committed 2 days ago

Verified3bff81d

<>

Tip

Update A2_S24_20220282_8.cpp

ISAAA7

committed 2 days ago

Verifieda407d4a

<>

Tip

Commits on Oct 30, 2023

Update A2_S24_20220282_5.cpp

ISAAA7

committed last week

Verifiedb91605d

<>

Tip

Commits on Oct 27, 2023

Question 2 implemented

ISAAA7

committed last week

Verifiedeccf5f4

<>

Tip

Add files via upload

ISAAA7

committed last week

Verifiedb0966ca

<>

Tip

Update A2_S23_20221119_9.cpp

ISAAA7 committed 45 minutes ago

Verified

1e9fc01

<>

Tip

Update and rename A2_S23_20221119_6(A,B).cpp to A2_S23_20221119_6.cpp

ISAAA7 committed 45 minutes ago

Verified

647cedc

<>

Tip

Update A2_S23_20221119_3.cpp

ISAAA7 committed 46 minutes ago

Verified

430d614

<>

Tip

Update and rename (reminder : kamel el comments ya jimmy)A2_S23_20221...

ISAAA7 committed 46 minutes ago

Verified

aef3185

<>

Tip

Commits on Oct 29, 2023

Update and rename A2_S23_20221119_12.cpp to (reminder : kamel el comm...

mogamal04 committed last week

Verified

8e5eb72

<>

Tip

Commits on Oct 28, 2023

Update A2_S23_20221119_9.cpp

mogamal04 committed last week

Verified

1d3e90c

<>

Tip

Update and rename A2_S23_20221119_6(A).cpp to A2_S23_20221119_6(A,B).cpp

mogamal04 committed last week

Verified

ccfbb8a

<>

Tip

Commits on Oct 27, 2023

Update and rename A2_S23_20221119_6.cpp to A2_S23_20221119_6(A).cpp

mogamal04 committed last week

Verified

8c3becb

<>

Tip

Update A2_S23_20221119_3.cpp

mogamal04 committed last week

Verified

d7bf619

<>

Tip

Add files via upload

mogamal04 committed last week

Verified

b890ca6

<>

Tip

Commits

History for OOP-Assignment-2 / A2_part1_S23_24_20221119_20220087_20220282 / A2_Task2_S23_24_20221119_20220087_20220282

Commits on Nov 5, 2023

Updating is valid function in BigReal.cpp

ISAAA7 committed 6 minutes ago

Verified

2b6cb6b

<>

Tip

UPDATING CONSTRUCTOR

Boleka3 committed 12 minutes ago

Verified

d879e3a

<>

Tip

Update BigReal.cpp

Boleka3 committed 13 minutes ago

Verified

0ba4b96

<>

Tip

Add files via upload

mogamal04 committed 41 minutes ago

Verified

13b6984

<>

Tip

End of commit history for this file

Newer Older

Work Break-Down Table

Names	IDs	Tasks
Belal Youness	20220087	(constructor, > , <)
Mohamed Gamal	20221119	(+, -, padding)
Mohamed Islam	20220282	(=, << , >>, is valid)

Algorithms

- **#include "BigReal.h":** This line includes the header file "BigReal.h," which contains the declarations and class definition for the BigReal class.
- **#include <regex>:** This line includes the C++ Standard Library header for regular expressions, which is used to validate input strings against a specific pattern.
- **BigReal::BigReal():** This is the implementation of the default constructor for the BigReal class. It initializes a BigReal instance with the default value of +0.0.
- **BigReal::BigReal(string real):** This is a constructor that takes a string as input and constructs a BigReal object from it. It checks the validity of the input string using a regular expression and extracts the sign, integer, and fractional parts.
- **BigReal::isValid(const string& real):** This is a member function of the BigReal class that checks if a given string represents a valid BigReal by applying a regular expression pattern.
- **BigReal::padding(BigReal &other):** This member function is a utility function used to ensure that two BigReal numbers have the same size, particularly in their integer and fractional parts. It adds leading zeros to align the parts for arithmetic operations.

- **BigReal::operator+(BigReal &other):** This member function overloads the addition operator for BigReal numbers. It handles addition between two BigReal instances, considering their signs, integer, and fractional parts.
- **BigReal::operator-(BigReal &other):** This member function overloads the subtraction operator for BigReal numbers. It handles subtraction between two BigReal instances, considering their signs, integer, and fractional parts.
- **BigReal::operator<(BigReal& nextReal):** This member function overloads the less than operator to compare two BigReal instances and determine which one is smaller. It takes into account both the integer and fractional parts.
- **BigReal::operator>(BigReal& nextReal):** This member function overloads the greater than operator to compare two BigReal instances and determine which one is greater. It considers both the integer and fractional parts.
- **BigReal::operator==(BigReal& nextReal):** This member function overloads the equality operator to check if two BigReal instances are equal in value. It considers the signs, integer, and fractional parts.

- **ostream &operator<<(ostream &COOUT, const BigReal & num):**
This is a global function that overloads the output stream operator (<<) to enable printing BigReal numbers to the console or other output streams in a user-friendly format. It takes a BigReal object as input and outputs it to the specified stream.
- **istream &operator>>(istream &in, BigReal &num):** This is a global function that overloads the input stream operator (>>) to allow reading and initializing BigReal objects from string input. It facilitates user-friendly input for BigReal numbers